

Application of Bayesian Techniques to Model the Burden of Human Salmonellosis Attributable to U.S. Food Commodities at the Point of Processing: Adaptation of a Danish Model

Mathematical models that estimate the proportion of foodborne illnesses attributable to food commodities at specific points in the food chain may be useful to risk managers and policy makers to formulate public health goals, prioritize interventions, and document the effectiveness of mitigations aimed at reducing illness. Using human surveillance data on laboratory-confirmed *Salmonella* infections from the Centers for Disease Control and Prevention and *Salmonella* testing data from U.S. Department of Agriculture Food Safety and Inspection Service's regulatory programs, we developed a point-of-processing foodborne illness attribution model by adapting the Hald *Salmonella* Bayesian source attribution model. Key model outputs include estimates of the relative proportions of domestically acquired sporadic human *Salmonella* infections resulting from contamination of raw meat, poultry, and egg products processed in the United States from 1998 through 2003. The current model estimates the relative contribution of chicken (48%), ground beef (28%), turkey (17%), egg products (6%), intact beef (1%), and pork (<1%) across 109 *Salmonella* serotypes found in food commodities at point of processing. While interpretation of the attribution estimates is constrained by data inputs, the adapted model shows promise and may serve as a basis for a common approach to attribution of human salmonellosis and food safety decision-making in more than one country.